

Common Identities

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1 Introduction

This section of notes does not follow the main flow of these lectures and notes. This is simply an organized directory of common identities and is NOT comprehensive.

Additionally, it serves similar to a reference sheet for equations but does not explain the mathematics involved in any of the equations. That is covered in the other notes and lectures.

2 Logarithmic and Exponential functions

Exponential

Let $x, y, b \in \mathbb{R}$

$$b^{x+y} = b^x b^y$$
$$b^{-x} = \frac{1}{b^x}$$

Logarithmic

Let $x, y, b \in \mathbb{R}$

$$\log_b(x) = y \iff b^y = x$$

$$\log_b(xy) = \log_b(x) + \log_b(y)$$

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

$$\log_b(x^y) = y \log_b(x)$$

Change of Basis

Let $a, b, c \in \mathbb{R}$

$$\log_b(a) = \frac{\log_c(a)}{\log_c(b)}$$

3 Trigonometric Identity's

All trigonometry will assume radians.

Pythagorean

$$\sin^2(x) + \cos^2(x) = 1$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$\tan^2(x) + 1 = \sec^2(x)$$

Reciprocal

$$\sin(x) = \frac{1}{\csc(x)} \quad \csc(x) = \frac{1}{\sin(x)}$$

$$\cos(x) = \frac{1}{\sec(x)} \quad \sec(x) = \frac{1}{\cos(x)}$$

$$\tan(x) = \frac{1}{\cot(x)} \quad \cot(x) = \frac{1}{\tan(x)}$$

Even Odd

Both sine and tangent are an odd functions and cosine is an even function resulting in the forms

$$\sin(-x) = -\sin(x)$$

$$\cos(-x) = \cos(x)$$

$$\tan(-x) = -\tan(x)$$

Additive

$$\sin(x \pm y) = \sin(x) \cos(y) \pm \cos(x) \sin(y)$$

$$\cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y)$$

$$\tan(x \pm y) = \frac{\tan(x) \pm \tan(y)}{1 \mp \tan(x) \tan(y)}$$

Double Angle

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$= 2 \cos^2(x) - 1$$

$$= 1 - 2 \sin^2(x)$$

$$\tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$$

Half Angle

$$\sin^2\left(\frac{x}{2}\right) = \frac{1 - \cos x}{2}$$

$$\cos^2\left(\frac{x}{2}\right) = \frac{1 + \cos x}{2}$$

$$\tan\left(\frac{x}{2}\right) = \frac{\sin(x)}{1 + \cos(x)} = \frac{1 - \cos(x)}{\sin(x)}$$

Cofunction

This assumes radians.

$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x) \quad \cos\left(\frac{\pi}{2} - x\right) = \sin(x)$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot(x) \quad \cot\left(\frac{\pi}{2} - x\right) = \tan(x)$$

$$\csc\left(\frac{\pi}{2} - x\right) = \sec(x) \quad \sec\left(\frac{\pi}{2} - x\right) = \csc(x)$$

4 Derivatives of One Variable

Power Rule

Let $n \in \mathbb{R}$

$$\frac{d}{dx} x^n = nx^{n-1}$$

linear combinations

Let $a, b \in \mathbb{R}$ and $f, g : \mathbb{R} \rightarrow \mathbb{R}$.

$$\frac{d}{dx} [af(x) + bg(x)] = a \frac{d}{dx} f(x) + b \frac{d}{dx} g(x)$$

Product Rule

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$.

$$\frac{d}{dx} f(x)g(x) = f'(x)g(x) + f(x)g'(x)$$

A quick derivation of the quotient rule from the product rule.

$$\begin{aligned} \frac{d}{dx} \frac{f(x)}{g(x)} &= \frac{d}{dx} f(x)[g(x)]^{-1} \\ &= f'(x)[g(x)]^{-1} + f(x)(-1[g(x)]^{-2}g'(x)) \\ &= \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{[g(x)]^2} \\ &= \frac{f'(x)}{g(x)} \frac{g(x)}{g(x)} - \frac{f(x)g'(x)}{[g(x)]^2} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2} \end{aligned}$$

Quotient Rule

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ and $g(x) \neq 0$

$$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Check this definition against analysis definition to confirm $g=0$ is not a problem

Chain Rule

Suppose $f, g : \mathbb{R} \rightarrow \mathbb{R}$.

$$\frac{d}{dx}(f \circ g)(x) = \frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

Trigonometric functions

Assuming radians

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

Inverse Trigonometric functions

Assuming radians

$$\frac{d}{dx} \arcsin(x) = \frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}, x \neq \pm 1$$

$$\frac{d}{dx} \arccos(x) = \frac{d}{dx} \cos^{-1}(x) = \frac{-1}{\sqrt{1-x^2}}, x \neq \pm 1$$

$$\frac{d}{dx} \arctan(x) = \frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \cot^{-1}(x) = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} \csc^{-1}(x) = \frac{-1}{|x|\sqrt{x^2-1}}, x \neq \pm 1, 0$$

$$\frac{d}{dx} \sec^{-1}(x) = \frac{1}{|x|\sqrt{x^2-1}}, x \neq \pm 1, 0$$

For the remainder of this section assume $u : \mathbb{R} \rightarrow \mathbb{R}$ is such that $u = u(x)$ meaning u is a function of x . Further if $u = x$ recall from the power rule $u' = 1$, thus the formulas all have the chain rule applied to them but if this is an unnecessary step in your analysis you can simply plug in $u = x$.

Exponential functions

Let $u : \mathbb{R} \rightarrow \mathbb{R}$.

$$\frac{d}{dx} b^u = \ln(b) b^u u'$$
$$\frac{d}{dx} e^u = e^u u'$$

Logarithmic functions

Let $u : \mathbb{R} \rightarrow \mathbb{R}$.

$$\frac{d}{dx} \log_b(u) = \frac{1}{u \ln(b)} u'$$
$$\frac{d}{dx} \ln(u) = \frac{1}{u} u'$$

5 Integrals and Anti-derivatives of One Variable

Due to the vast variability in integration, this section will outline integration by parts and point to a table of integrals for the remainder of common integrals.

Integration by Parts

Let $u, v : \mathbb{R} \rightarrow \mathbb{R}$.

$$\int u dv = uv - \int v du$$